ELEC40010 - MATHEMATICS1A - SOLUTIONS

Section A: Multiple Choice

Enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks. Total marks for this section: 40.

1. In complex exponential form, $re^{i\theta}$, the complex number $\dfrac{i-1}{i+1}$ has

   (a) $r = 1/2$ and $\theta = \pi$, (b) $r = 1/2$ and $\theta = \pi/2$,

   (c) $r = 1$ and $\theta = -\pi/2$, (d) $r = 1$ and $\theta = \pi/2$,

   (e) none of the above [3]

   Solution

   $$\frac{-1+i}{1+i}\left(\frac{1-i}{1-i}\right) = \frac{2i}{2} = i = 1e^{i\pi/2}$$

   so (d)

2. The roots of the equation $z^3+z^2+z+1=0$ are

   (a) one real root, and two complex conjugate roots ,

   (b) two real roots, and one complex root ,

   (c) unobtainable , (d) three real roots ,

   (e) none of the above [3]

   Solution Spot that $-1+1-1+1=0$ so $z=-1$ is a solution and can factorize

   $$z^3+z^2+z+1 = (z+1)(z^2+1)$$

   The quadratic has roots $\pm i$, so (a). Alternatively, differentiate: $3x^2+2x+1=0$ has no real roots, so no stationary points, curve crosses $x-$axis only once, at the single real root.

3. Three functions are given by

   $$(I)\ x\cos\left(\frac{\pi}{2}-x\right) , \quad (II)\ \sin[\ln(1+x^2)] , \quad (III)\ \frac{\cos x}{x} .$$

   Assuming the domains are defined for each function, which of the following statements is **true**?

   (a) (II) is not periodic and (I) is even , (b) (II) is periodic and (III) is odd ,

   (c) (III) is periodic and (II) is not , (d) (I) and (III) are both even ,

(e) none of the above. [3]

Solution (I) is $x\sin x$, so even, (II) is even, (III) is odd. None are periodic. so (a)

4. The limit: $\lim_{x\to 0} \dfrac{x^2}{1-x-e^{-x}}$, exists and is equal to

(a) 1 , (b) -2 , (c) 3 , (d) -4 ,

(e) none of the above. [3]

Solution It's "$0/0$" so try l'Hopital

$$\lim_{x\to 0} \frac{x^2}{1-x-e^{-x}} = \lim_{x\to 0} \frac{2x}{-1+e^{-x}}$$

Still "$0/0$" so l'Hopital again:

$$= \lim_{x\to 0} \frac{2}{-e^{-x}} = -2$$

so (b).

5. The function $y = \dfrac{x^2-x+1}{x-1}$ has

(a) a minimum at $x=0$, (b) a maximum at $x=2$,

(c) a point of inflection in the interval $[2,\infty)$, (d) a diagonal asymptote ,

(e) none of the above. [3]

Solution See Sect B, Q2a. (d)

6. Given $x+y-\ln(x-y)=42$, which of the following is the derivative $\dfrac{dy}{dx}$?

(a) $\dfrac{x+y-1}{x-y+1}$, (b) $-\dfrac{y-x+1}{x-y-1}$, (c) $-\dfrac{y-x+1}{y-x-1}$, (d) $\dfrac{x-y-1}{x+y+1}$,

(e) none of the above. [3]

Solution Differentiate implicitly:

$$1+\frac{dy}{dx}-\frac{1}{x-y}\left(1-\frac{dy}{dx}\right)=0 \Rightarrow \left(1+\frac{1}{x-y}\right)\frac{dy}{dx}=\frac{1}{x-y}-1$$

Tidy up, answer: (c).

7. The integral $I = \int \frac{1}{1+\cos x}\, dx$ is equal to

(a) $\tan(x/2)+\mathrm{C}$, (b) $\cot x - \csc x + \mathrm{C}$, (c) $\frac{x}{1-\sin x}+\mathrm{C}$, (d) $-\cot(x/2)+\mathrm{C}$,

(e) none of the above [3]

Solution Use substitution $t = \tan(x/2)$, with $dx = \frac{2}{1+t^2}\, dt$ and $\cos x = \frac{1-t^2}{1+t^2}$, from formula sheet to rewrite integral:

$$I = \int \frac{1}{1+\frac{1-t^2}{1+t^2}} \frac{2}{1+t^2}\, dt \;=\; \int \frac{2}{1+t^2+1-t^2}\, dt \;=\; t+C = \tan(x/2)+C$$

so (a).

8. The integral $\int \frac{x(x-1)}{(x^2+1)(x+1)}\, dx$ is equal to

(a) $\ln(x^2+1) - \ln(x+1) + \mathrm{C}$, (b) $\ln(x+1) - \tan^{-1}(x^2+1) + \mathrm{C}$,

(c) $\frac{2}{(x+1)^2} - \ln(x+1) + \mathrm{C}$, (d) $\ln(x+1) - \tan^{-1} x + \mathrm{C}$,

(e) none of the above [3]

Solution Partial fractions:

$$\frac{x(x-1)}{(x^2+1)(x+1)} \;=\; \frac{1}{x+1} - \frac{1}{x^2+1}$$

so (d)

9. The power series

$$\sum_{n=1}^{\infty} \frac{(x-1)^n}{\sqrt[3]{n}}$$

has radius of convergence R and interval of convergence I, where

(a) $R = 1$ and $I = (0,2)$, (b) $R = 0$ and $I = [1,2)$,

(c) $R = 1$ and $I = (0,2]$, (d) $R = 2$ and $I = [-1,0)$,

(e) none of the above [3]

Solution Ratio test:

$$\lim_{n\to\infty} \left| \frac{\frac{(x-1)^{n+1}}{\sqrt[3]{n+1}}}{\frac{(x-1)^n}{\sqrt[3]{n}}} \right| \;=\; \lim_{n\to\infty} \sqrt[3]{\frac{n}{n+1}} |x-1| \;=\; |x-1| < 1 \Rightarrow -1 < x-1 < 1$$

for convergence. So $R = 1$ and I includes the open interval $(0,2)$. Test at endpoints: $x = 0$ gives $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}}$ which converges, and $x = 2$ gives $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}}$ which diverges. Hence $I = [0,2)$, so (e).

10. Given the linear first-order ordinary differential equation (ODE)

$$t\frac{dx}{dt} + 2x = t^2,$$

which of the following is an integrating factor?

(a) x^2, (b) $\frac{1}{t}$, (c) t^2, (d) e^{2t},

(e) none of the above. [2]

Solution Write in standard form:

$$\frac{dx}{dt} + \frac{2}{t}x = t,$$

so integrating factor is $e^{\int \frac{2}{t}\,dt} = t^2$, so **answer (c)**

11. The solution of the linear ODE

$$\frac{dy}{dx} + \frac{2}{x}y = x$$

is given by

(a) $y = \frac{1}{2}x^2 + C$, (b) $x = \frac{C}{t^2} - \frac{1}{4}t^2$, (c) $y = x^2\left(x + \frac{C}{x}\right)$, (d) $y = \frac{x^2}{4} + \frac{C}{x^2}$,

(e) none of the above. [2]

Solution Same ODE as Q10. With I.F x^2, we rewrite ODE as

$$\frac{d}{dx}\left(x^2 y\right) = x^2\left(\frac{dy}{dx} + \frac{2}{x}y\right) = x^3 \Rightarrow x^2 y = \int x^3\,dx = \frac{1}{4}x^4 + C \Rightarrow$$

Answer: (d)

12. Given the linear second-order ODE

$$4\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + x = 25\sin t,$$

which of the following is the complementary function?

(a) $x_c = c_1 t e^{\frac{1}{2}t} + c_2 e^{\frac{1}{2}t}$, (b) $x_c = c_1 e^{-\frac{1}{2}t} + c_2 e^{\frac{1}{2}t}$,

(c) $x_c = e^{\frac{1}{2}t}(c_1\cos 2t + c_2\sin 2t)$, (d) $y_c = c_2 e^{-2x} + c_2 e^{2x}$,

(e) none of the above. [2]

Solution Auxiliary equation

$$4\lambda^2 - 4\lambda + 1 = 0 \Rightarrow \lambda = \frac{1}{2}$$

repeated root, hence answer is (a).

13. Given the linear second-order ODE

$$4\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + x = 25\sin t\,,$$

which of the following is a particular integral?

(a) $x_p = -3\cos t + 4\sin t\,,$ (b) $x_p = 4e^t \sin(t+\pi/3),,$

(c) $y_p = -3\cos x + 4\sin x\,,$ (d) $x_p = 4\cos t - 3\sin t$

(e) none of the above. [2]

Solution Try

$$y_p = a\cos t + b\sin t \Rightarrow y_p' = -a\sin t + b\cos t \Rightarrow y_p'' = -a\cos t - b\sin t$$

substitute into the ODE:
$4(-a\cos t - b\sin t) - 4(-a\sin t + b\cos t) + a\cos t + b\sin t$

$$= \ (-4a - 4b + a)\cos t + (-4b + 4a + b)\sin t \ = \ 25\sin t$$

and equate coefficients : $-3a - 4b = 0$ and $-3b + 4a = 25$ solve for a, b to get **Answer: (d)** .

14. Given the linear second-order ODE

$$4\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + x = 25\sin t\,,$$

satisfying initial conditions $x(0) = 4$ and $x'(0) = -2$, which of the following is the solution?

(a) $y = e^{-2x} - e^{2x} - 3\cos x + 4\sin x$, (b) $x = e^{-\frac{1}{2}t} - e^{\frac{1}{2}t} + 4e^t \sin(t+\pi/3)$,

(c) $x = te^{\frac{1}{2}t} + 4\cos t - 3\sin t$, (d) $x = te^{\frac{1}{2}t} - e^{\frac{1}{2}t} - 3\cos t + 4\sin t$,

(e) none of the above. [2]

Solution Putting together solution from Q12 and Q13, we have

$$x(t) = c_1 te^{\frac{1}{2}t} + c_2 e^{\frac{1}{2}t} + 4\cos t - 3\sin t$$

so

$$x'(t) = \left(1 + \frac{1}{2}t\right) c_1 e^{\frac{1}{2}t} + \frac{1}{2}c_2 e^{\frac{1}{2}t} - 4\sin t - 3\cos t$$

With initial conditions we have

$$x(0) = c_2 + 4 = 4 \Rightarrow c_2 = 0$$

and

$$x'(0) = c_1 - 3 = -2 \Rightarrow c_1 = 1,$$

so (c).

15. Let a function of two variables be given as

$$f(x,y) = \tan^{-1}\left(\frac{x}{y}\right) .$$

The first- and second-order partial derivatives with respect to x are

(a) $\dfrac{\partial f}{\partial x} = \dfrac{y}{x^2+y^2}$, and $\dfrac{\partial^2 f}{\partial x^2} = \dfrac{2xy}{(x^2+y^2)^2}$,

(b) $\dfrac{\partial f}{\partial x} = \dfrac{x}{x^2+y^2}$, and $\dfrac{\partial^2 f}{\partial x^2} = -\dfrac{2xy}{(x^2+y^2)^2}$,

(c) $\dfrac{\partial f}{\partial x} = -\dfrac{x}{x^2+y^2}$, and $\dfrac{\partial^2 f}{\partial x^2} = \dfrac{2xy}{(x^2+y^2)^2}$,

(d) $\dfrac{\partial f}{\partial x} = \dfrac{y}{x^2+y^2}$, and $\dfrac{\partial^2 f}{\partial x^2} = -\dfrac{2xy}{(x^2+y^2)^2}$,

(e) none of the above. [3]

Solution

$$\frac{\partial}{\partial x}\left[\tan^{-1}\left(\frac{x}{y}\right)\right] = \frac{1}{1+\left(\frac{x}{y}\right)^2}\frac{\partial}{\partial x}\left(\frac{x}{y}\right) = \frac{1}{1+\left(\frac{x}{y}\right)^2}\left(\frac{1}{y}\right) = \frac{y}{x^2+y^2}$$

and again, for the second derivative,

$$\frac{\partial}{\partial x}\left[\frac{y}{x^2+y^2}\right] = -\frac{y}{(x^2+y^2)^2}(2x),$$

so (d).

Section B

Enter your full answers in the answer booklet. This section has three questions, each worth 20 marks.

1. a) Find all complex solutions $z = x + iy$, of the equation

$$\sin(iz) = \frac{4}{3} .$$

[7]

Solution: Begin with

$$\sin[i(x+iy)] = \sin(ix-y) = \sin(ix)\cos y - \cos(ix)\sin y = i\sinh x\cos y - \cosh x\sin y = \tfrac{4}{3}$$

Equating real and imaginary parts we have two equations:

$$(i)\ \sinh x\cos y = 0 \quad \text{and} \quad (ii)\ \cosh x\sin y = -\frac{4}{3}$$

The first equation gives

$$\sinh x = 0 \Rightarrow x = 0\,, \quad \text{or} \quad \cos y = 0 \Rightarrow y = (2n+1)\frac{\pi}{2}\ ,$$

for integers n. First, substitute $x = 0$ into (ii), and get

$$\sin y = -\frac{4}{3}$$

which has no solutions.
The only possible solutions will have $y = (2n+1)\pi/2$. Substituting this into (ii) we have

$$\cosh x \sin(2n+1)\frac{\pi}{2} = -\frac{4}{3} \Rightarrow \cosh x = \pm\frac{4}{3}$$

As $\cosh x > 0$, we choose $\cosh x = 4/3 \Rightarrow x = \pm\cosh^{-1}(4/3)$ corresponding to $\sin(2n+1)\pi/2 = 1 \Rightarrow n$ is even. Finally we have the solutions

$$z = \pm\cosh^{-1}\left(\frac{4}{3}\right) + i\left(\frac{\pi}{2} + 2k\pi\right)\ ,$$

for integers k.

b) Obtain the limits [7]

$$(i)\ \lim_{x\to -1}\frac{2-\sqrt{-4x}}{x+1} \quad \text{and} \quad (ii)\ \lim_{x\to\infty}\sqrt[3]{3x}\left[(3x-1)^{2/3}-(3x+1)^{2/3}\right].$$

Solution: (i)

$$\lim_{x\to -1}\frac{(2-\sqrt{-4x})}{(x+1)}\frac{(2+\sqrt{-4x})}{(2+\sqrt{-4x})} = \lim_{x\to -1}\frac{4-(-4x)}{(x+1)(2+\sqrt{-4x})} = \lim_{x\to -1}\frac{4}{2+\sqrt{-4x}} = \frac{4}{2+2} = 1\,.$$

(ii) Rewrite to enable binomial expansion:

$$(3x\pm 1)^{2/3} = (3x)^{2/3}\left(1\pm\frac{1}{3x}\right)^{2/3} = (3x)^{2/3}\left[1+\frac{2}{3}\left(\pm\frac{1}{3x}\right)+\frac{\frac{2}{3}(\frac{2}{3}-1)}{2}\left(\pm\frac{1}{3x}\right)^2+\ldots\right]$$

Hence

$$\sqrt[3]{3x}\left[(3x-1)^{2/3}-(3x+1)^{2/3}\right] = (3x)^{1/3}(3x)^{2/3}\left[1-\frac{2}{9x}-\frac{1}{81x^2}+\cdots-\left(1+\frac{2}{9x}-\frac{1}{81x^2}+\cdots\right)\right]$$

and even powers cancel, leaving

$$= 3x\left(-\frac{4}{9x}+\frac{k_1}{x^3}+\frac{k_2}{x^5}+\ldots\right) = 3\left(-\frac{4}{9}+\frac{k_1}{x^2}+\frac{k_2}{x^4}+\cdots\right)$$

for constants k_i and in the limit $x\to\infty$, all terms with x in denominator vanish, leaving only $-4/3$.

c) For a function $y(x)$ where both y and x are given in terms of a parameter s, in the form

$$y = y(s) \quad \text{and} \quad x = x(s)\,,$$

show that

$$\frac{dy}{dx} = \frac{\frac{dy}{ds}}{\frac{dx}{ds}}$$

and obtain

$$\frac{d^2y}{dx^2}$$

in terms of derivatives of x and y with respect to s. [6]

Solution:

$$\frac{dy}{dx} = \frac{d}{dx}[y(s)] = \frac{d}{ds}(y(s))\frac{ds}{dx} = \frac{dy}{ds}\frac{ds}{dx} = \frac{\frac{dy}{ds}}{\frac{dx}{ds}}\,.$$

using chain and inverse function rules. Now differentiate again, using chain rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{\frac{dy}{ds}}{\frac{dx}{ds}}\right) = \frac{d}{ds}\left(\frac{\frac{dy}{ds}}{\frac{dx}{ds}}\right)\frac{ds}{dx} = \left(\frac{\frac{d}{ds}\left(\frac{dy}{ds}\right)\frac{dx}{ds}-\frac{dy}{ds}\frac{d}{ds}\left(\frac{dx}{ds}\right)}{\left(\frac{dx}{ds}\right)^2}\right)\frac{ds}{dx} = \frac{\frac{d^2y}{ds^2}\frac{dx}{ds}-\frac{d^2x}{ds^2}\frac{dy}{ds}}{\left(\frac{dx}{ds}\right)^3}$$

using the quotient and inverse function rules in the second and third step.

2. a) Given the function $f(x) = \dfrac{x^2 - x + 1}{x - 1}$, obtain all stationary points and use the second derivative to classify them. Obtain all points of inflection and asymptotes and sketch the function, explaining your results. You may use results from section A. [6]

Solution: Rewrite as

$$f(x) = x + \frac{1}{x-1}$$

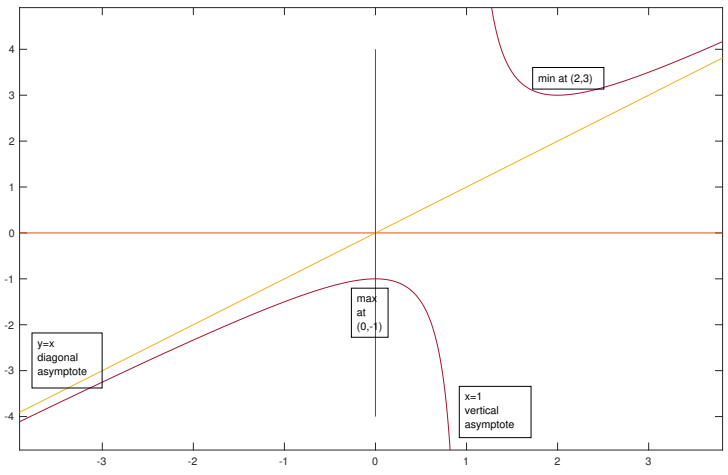
and differentiate twice:

$$f'(x) = 1 - \frac{1}{(x-1)^2} \Rightarrow f''(x) = \frac{2}{(x-1)^3}$$

Set $f'(x) = 0 \Rightarrow x = 0, 2$, so $(0,-1)$ and $(2,3)$ are stationary points. Evaluate the second derivative:

$f''(0) = -2 < 0$, so $(0,-1)$ is a maximum, and $f''(2) = 2 > 0$, so $(2,3)$ is a minimum.

As $x \to \pm\infty$, $1/(x-1) \to 0$, so $f(x) = x$ is a diagonal asymptote. As $x \to 1$, $f(x) - \to \pm\infty$, so $x = 1$ is a vertical asymptote. Given $f''(x) = 0$ has no solution, there are no points of inflection. This simplifies things. At the minimum $(2,3)$, $f(x) > x$, so curve lying above the diagonal asymptote. Hence to the right, the curve must approach the asymptote from above: as $x \to +\infty$, we have $f(x) \to x^+$. Likewise, as we approach the vertical asymptote from the right, the curve must become unbounded in the positive direction: as $x \to 1^+$, we have $f(x) \to +\infty$. A similar argument applies on the other side and we can sketch:



b) Obtain the indefinite integral $I = \int \sin^{-1} x \, dx$. Hence, or otherwise, determine the area enclosed by the line $y = \dfrac{\pi}{2}$, the curve $y = \sin^{-1} x$, and the $y-$axis. [7]

Solution: Integrate by parts: let $u = \sin^{-1} x \Rightarrow du = \frac{1}{\sqrt{1-x^2}}\, dx$, using the Formula sheet, and $dv = dx \Rightarrow v = x$. Hence

$$I \;=\; x \sin^{-x} x - \int \frac{x}{\sqrt{1-x^2}}\, dx \,.$$

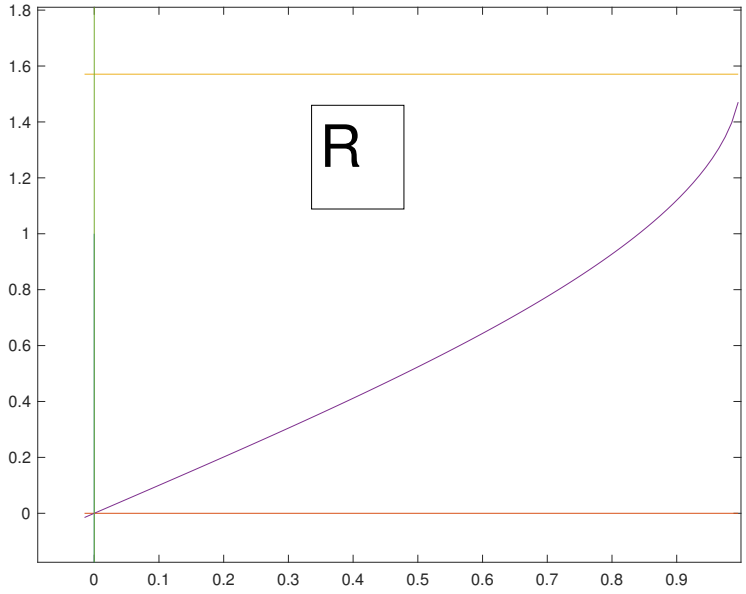
For the last integral substitute $u = 1 - x^2 \Rightarrow du = -2x dx$, so that

$$\int \frac{x}{\sqrt{1-x^2}}\, dx \;=\; -\frac{1}{2}\int u^{-1/2}\, du = -u^{1/2} = -\sqrt{1-x^2}$$

and we have

$$I \;=\; x \sin^{-x} x + \sqrt{1-x^2} + C \,.$$

The area in question is shown as the region R: the part of a rectangle of area $\pi/2$, lying above the curve $y = \sin^{-1} x$:



The part of the rectangle not included is below the curve $y = \sin^{-1} x$ and hence the area of R is

$$\frac{\pi}{2} - \int_0^1 \sin^{-1} x \, dx \;=\; \frac{\pi}{2} - \Big[x \sin^{-x} x + \sqrt{1-x^2} \Big]_0^1 \;=\; \frac{\pi}{2} - \big(\sin^{-1}(1) - 1 \big) \;=\; 1 \,.$$

c) Obtain the Maclaurin Series for $\cos x$ up to the term in x^4. Use this series to obtain an estimate for $\cos 1$, and use the Lagrange form of the remainder term to give the best bound you can obtain for the error incurred. [7]

Solution: Tabulate:

$$\begin{array}{rll} f(x) & = \cos x, & f(0) = 1 \\ f'(x) & = -\sin x, & f'(0) = 0 \\ f''(x) & = -\cos x, & f''(0) = -1 \\ f'''(x) & = \sin x, & f'''(0) = 0 \\ f^{(4)}(x) & = \cos x, & f^{(4)}(0) = 1 \\ f^{(5)}(x) & = -\sin x, & f^{(5)}(0) = 0 \\ f^{(6)}(x) & = -\cos x, & \end{array}$$

Hence the series is

$$f(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + R_4 = 1 - \frac{1}{2} + \frac{1}{24} + R_4$$

and we estimate $\cos 1 = 13/24$. The remainder R_4 gives a bound for the error, but this can be improved, noting that expanding to x^5 adds no new terms to the series, but the remainder is

$$R_5 = -\frac{\cos c}{6!} x^6$$

with $0 < c < 1$. Cosine is decreasing on $[0,1]$ so $|\cos c| < 1$ giving a bound for the error of

$$|R_5| < \frac{1}{6!} 1^6 = \frac{1}{6!}$$

3. a) Solve the first-order ODE

$$\frac{dy}{dx} = \frac{x+y}{2x-y} + \frac{3y}{2x}$$

satisfying the initial condition $y(1) = 2$. [7]

Solution: This is a homogenous first-order ODE:

$$\frac{dy}{dx} = \frac{1+\frac{y}{x}}{2-\frac{y}{x}} + \frac{3}{2}\left(\frac{y}{x}\right)$$

and substituting $v = \frac{y}{x} \Rightarrow \frac{dy}{dx} = x\frac{dv}{dx} + v$ we have

$$x\frac{dv}{dx} + v = \frac{1+v}{2-v} + \frac{3}{2}v \Rightarrow x\frac{dv}{dx} = \frac{1+v}{2-v} + \frac{1}{2}v = \frac{-\frac{1}{2}v^2+2v+1}{-v+2}$$

so that

$$\int \frac{-v+2}{-\frac{1}{2}v^2+2v+1}\,dv = \int \frac{1}{x}\,dx \Rightarrow \ln\left(-\frac{1}{2}v^2+2v+1\right) = \ln(x)+C \Rightarrow -\frac{1}{2}\left(\frac{y}{x}\right)^2+2\left(\frac{y}{x}\right)+1 = kx$$

for some k. With the initial condition $y(1) = 2$ we have

$$-\frac{1}{2}(2^2)+2(2)+1 = k \Rightarrow k = 3$$

giving the solution

$$-\frac{1}{2}\left(\frac{y}{x}\right)^2+2\left(\frac{y}{x}\right)+1 = 3x\,.$$

b) Use Leibnitz' theorem to differentiate the second-order ODE

$$\frac{d^2y}{dx^2} - x\frac{dy}{dx} + xy = 0$$

n times, and let $x = 0$ to obtain the recurrence relation

$$y^{(n+2)}(0) = ny^{(n)}(0) - ny^{(n-1)}(0) \ , \qquad (n \geq 1)$$

where $y^{(k)}(0)$ is $y(x)$ differentiated k times, and evaluated at $x = 0$. Hence obtain the first five non-zero terms of the Maclaurin series of $y(x)$, given that $y(0) = 1$ and $y'(0) = -1$. [8]

Solution: Differentiate using Leibnitz:

$$\frac{d^n}{dx^n}(xy'(x) = xy^{(n+1)}(x) + ny^{(n)}(x)$$

and similarly for xy, so differentiating the ODE n times, we get

$$y^{(n+2)}(x) - xy^{(n+1)}(x) - ny^{(n)}(x) + xy^{(n)}(x) + ny^{(n-1)}(x) = 0 \ ,$$

and let $x = 0$:

$$y^{(n+2)}(0) - ny^{(n)}(0) + ny^{(n-1)}(0) = 0 \Rightarrow y^{(n+2)}(0) = ny^{(n)}(0) - ny^{(n-1)}(0) \ ,$$

as required.

To use the initial conditions and the recurrence, we need $y''(0) = 0$, from the ODE itself, so that
$y^{(0)}(0) = y(0) = 1$,
$y^{(1)}(0) = y'(0) = -1$,
$y^{(2)}(0) = y''(0) = 0$,
now let $n = 1$ in the recurrence relation:
$y^{(3)}(0) = y^{(1+2)}(0) = 1y^{(1)}(0) - 1y^{(1-1)}(0) = y'(0) - y(0) = -2$,
let $n = 2$: $y^{(4)}(0) = 2y^{(2)}(0) - 2y^{(1)}(0) = 2(0) - 2(-1) = 2$,
let $n = 3$: $y^{(5)}(0) = 3y^{(3)}(0) - 3y^{(2)}(0) = 3(-2) - 3(0) = -6$.
This is enough for the series:
$y(x) = y(0) + y'(0)x + \dfrac{y''(0)}{2!}x^2 + \cdots$ so the first five non-zero terms are

$$y(x) \approx 1 - x + \frac{-2}{3!}x^3 + \frac{2}{4!}x^4 + \frac{-6}{5!}x^5 = 1 - x - \frac{1}{3}x^3 + \frac{1}{12}x^4 - \frac{1}{20}x^5$$

c) Let a function of two variables be given as

$$f(x,y) = \tan^{-1}\left(\frac{x}{y}\right) .$$

Obtain the first- and second-order partial derivatives with respect to y. Is $f(x,y)$ a solution of Laplace's equation, $\nabla^2 f = 0$? You may use results from section A. [5]

Solution: In A, Q15, we found that

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = -\frac{2xy}{(x^2+y^2)^2} .$$

Similarly,

$$\frac{\partial}{\partial y}\left[\tan^{-1}\left(\frac{x}{y}\right)\right] = \frac{1}{1+\left(\frac{x}{y}\right)^2}\frac{\partial}{\partial y}\left(\frac{x}{y}\right) = \frac{1}{1+\left(\frac{x}{y}\right)^2}\left(-\frac{x}{y^2}\right) = -\frac{x}{x^2+y^2}$$

and again, for the second derivative,

$$f_{yy} = \frac{\partial}{\partial y}\left[-\frac{x}{x^2+y^2}\right] = \frac{x}{(x^2+y^2)^2}(2y) = \frac{2xy}{(x^2+y^2)^2}$$

Hence $\nabla^2 f = f_{xx} + f_{yy} = -\frac{2xy}{(x^2+y^2)^2} + \frac{2xy}{(x^2+y^2)^2} = 0$, so yes, f is a solution of Laplace's equation.